Chapter 7. Logit Models for Multinomial Responses

Baseline-category and cumulative logits

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7.1 Nominal responses: baseline-category logit models

Motivation. Many responses have more than two unordered categories (food type, brand, party). Pairing each category with a baseline gives $J-1$ simultaneous logit equations that together describe all pairwise odds, extending binary logistic regression.

Let Y have J categories and $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$. At a fixed setting $\mathbf{x}$, the counts in the J categories are multinomial with probabilities $\{\pi_1(\mathbf{x}), \ldots, \pi_J(\mathbf{x})\}$. Of the $\binom{J}{2}$ pairwise logits, any suitable $J-1$ determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with the convention $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. Hence the choice of baseline is arbitrary: changing it reparametrizes but does not change the model or its fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention makes the formula valid when a or b equals J). The set of probability vectors described is the same for every baseline, since each parametrization is a linear reparametrization of the others. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; with continuous predictors or sparse data they remain valid for comparing nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

For 219 alligators from four Florida lakes, the response is the primary food type by volume (fish, invertebrate, reptile, bird, other); predictors are lake L, gender G and size S (≤ 2.3 m vs. > 2.3 m). The data are sparse (219 observations in 80 cells), so G^2 is more reliable for comparing models than for testing fit. Comparisons give $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$, each on $df = 4$, so we collapse over gender. For the collapsed table, $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), whereas (L) has 38.2 and (S) 66.2; both L and S matter. Observed counts (fitted values for $L+S$ in parentheses):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Only two of the 40 standardized residuals exceed 2 in absolute value. With fish as baseline, dummies $s = 1$ for small size and z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George is the reference), the ML estimates (SE) are:

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have estimated odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large alligators (Wald 95% CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$). Those odds are relatively higher in Trafford and Oklawaha and lower in Hancock than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

The parameters of the baseline category are set to zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$, $j = 1, \ldots, J$, leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and similarly $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. With a quantitative predictor (actual length), plotting $\hat{\pi}_j(x)$ against x is informative; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone in x.

7.2 Ordinal responses: cumulative logit models

Motivation. When the categories are ordered, a model that uses the ordering is more parsimonious and has more power (Section 6.4). Cumulative logits with a common effect across cutpoints give the most popular such model, motivated by an underlying continuous latent response.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; a model for one of them alone is a binary logit model for the collapsing $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.5 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^β is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every collapsing of Y into $(\le j, > j)$: hence McCullagh's name *proportional odds*. The response curves for $j = 1, 2, 3$ ($J = 4$) have exactly the same shape (Figure 1).

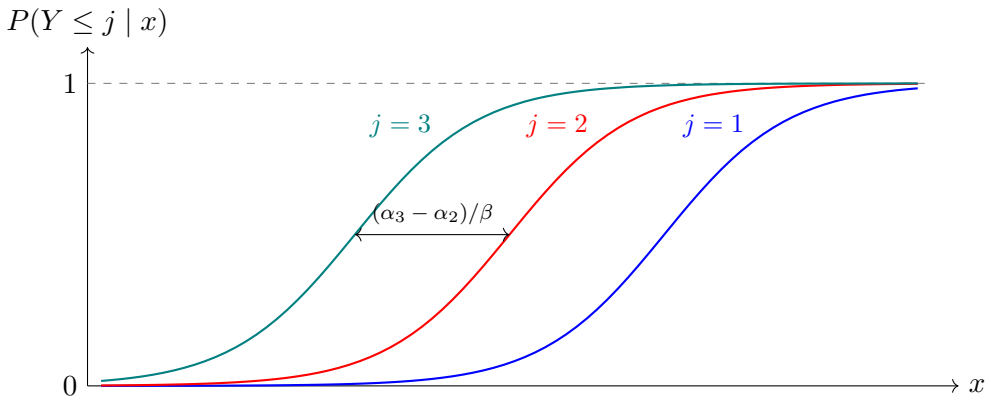


Figure 1: Proportional odds model with $J = 4$ and $\beta > 0$: the three cumulative probability curves are logistic curves with a common slope, horizontally displaced from one another.

Because the $J - 1$ curves are constrained to share a shape, the fit differs from separate binary fits. With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n} \prod_{j=1}^{J} \left[\frac{\exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)} - \frac{\exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)} \right]^{y_{ij}} \tag{7.4}$$

with $\alpha_0 = -\infty$, $\alpha_J = +\infty$; it is maximized by Fisher scoring.

7.2.3 Mental impairment example

Example 7.6 (Mental impairment)

A random sample of 40 adults in Alachua County, Florida. Y = mental impairment (1 well, 2 mild symptom formation, 3 moderate symptom formation, 4 impaired); x_1 = life events index (number and severity of important events in the past three years); x_2 = SES (1 high, 0 low). Listed as (x_2, x_1) within each response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	-0.282	0.642	$(-1.56,\ 0.98)$
α_2	1.213	0.661	$(-0.05,\ 2.57)$
α_3	2.209	0.721	$(0.86,\ 3.71)$
life (β_1)	-0.319	0.121	$(-0.57,\ -0.09)$
SES (β_2)	1.111	0.611	$(-0.06,\ 2.35)$

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y=1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1+\exp[-0.282 - 0.319(4.275)]} = 0.16.$$

The cumulative probability from the "well" end decreases with life events and is higher at high SES: given life events, the odds of impairment below any fixed level at high SES are $e^{1.111} = 3.0$ times those at low SES. In probability terms, at $\bar{x}_1$, $\hat{P}(Y=1)$ is 0.37 (high SES) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high SES) and from 0.28 to 0.09 (low SES). The score test of proportional odds (separate β's for each logit, 4 extra parameters) gives 2.33, $df = 4$, $P = 0.68$.

7.2.4 More complex models

Interactions, factors and so on enter as in binary logistic regression; only the intercept is replaced by $J-1$ cutpoint intercepts. For the mental impairment data, adding the interaction gives

$$\text{logit}\,\hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which necessarily cross for some $\mathbf{x}$, violating the ordering of cumulative probabilities. A small P-value in the proportional odds test may reflect a large n more than important differences among the $\hat{\boldsymbol{\beta}}_j$, and the simpler model may have smaller MSE even if biased. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters;
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument works for any error distribution, so the logit can be replaced by other inverse cdfs. Probit and complementary log-log links give cumulative models with their own interpretations, and a scale component handles dispersion differences.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

Model (7.5) arises from crude measurement of a latent Y^* following a location family $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on a standardized latent scale (using $-\boldsymbol{\beta}$ instead of $+\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$): appropriate when Y^* is normal. In $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k standard deviations of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, from an extreme-value latent distribution): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$. It is the grouped-time version of the proportional hazards model and is sometimes called a *proportional hazards* model. $P(Y \le j)$ approaches 1 faster than it approaches 0. The log-log link suits the case where cloglog holds for the reversed categories (reversing the order of the categories turns G into $1 - G(-z)$).

Proposition 7.7 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp\left[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}\right]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs. McCullagh's Fisher scoring algorithm uses the likelihood written as in (7.4) with G in place of the logistic cdf; for large n the maximum is unique, and the log likelihood is concave for the logit, probit and complementary log-log links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.8 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ages, suggesting the cloglog link (also implied by an exponentially increasing hazard, the Gompertz extreme-value law). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat{\beta}_1 = -0.658,\ \hat{\beta}_2 = 0.626.$$

(Fit statistics are irrelevant: these are population values.) By Proposition 7.7, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival proportion beyond any age is women's raised to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. If white women survive beyond some age with probability π, white men and black women do so with probability about π^2, and black men with about π^4.

7.4 Selected exercises

Motivation. These theory problems from the end of the chapter in Agresti fill in structural facts used above: why baseline-category logits are the natural parameters, how a single category probability can be nonmonotone, why cumulative logits need a common effect, and how the proportional odds model describes association in a two-way table.

Exercise 7.24 (Multinomial trial as an exponential family)

For a single multinomial trial with probabilities $\pi_1, \dots, \pi_J$ (all positive), let $\mathbf{y} = (y_1, \dots, y_{J-1})^\top$ with $y_j = 1$ if the outcome is category j and 0 otherwise. Show that the distribution of $\mathbf{y}$ has the form

$$f(\mathbf{y}; \boldsymbol{\theta}) = \exp\big[\mathbf{y}^\top \boldsymbol{\theta} - b(\boldsymbol{\theta})\big],$$

a $(J-1)$-parameter exponential family whose natural parameters are the baseline-category logits $\theta_j = \log(\pi_j/\pi_J)$. Find b and check that $\partial b/\partial \theta_j = \mathbb{E}(y_j)$.

Solution.

Step 1. Put $y_J = 1 - \sum_{j<J} y_j$. Exactly one of $y_1, \dots, y_J$ equals 1, so $P(\mathbf{y}) = \prod_{j=1}^{J} \pi_j^{y_j}$.

Step 2. Factor out π_J:

$$\prod_{j=1}^{J} \pi_j^{y_j} = \pi_J^{1-\sum_{j<J} y_j} \prod_{j<J} \pi_j^{y_j} = \pi_J \prod_{j<J} \Big(\frac{\pi_j}{\pi_J}\Big)^{y_j} = \exp\Big[\sum_{j<J} y_j \theta_j + \log \pi_J\Big].$$

Step 3. By Proposition 7.3 (with no predictors), $\pi_J = 1/(1 + \sum_{h<J} e^{\theta_h})$, so $-\log \pi_J = b(\boldsymbol{\theta}) := \log(1 + \sum_{h<J} e^{\theta_h})$, a function of $\boldsymbol{\theta}$ alone. This gives the stated form with dispersion $a(\phi) = 1$ and $c(\mathbf{y}, \phi) = 0$; the map $(\pi_1, \dots, \pi_{J-1}) \mapsto \boldsymbol{\theta}$ is one-to-one onto $\mathbb{R}^{J-1}$.

Step 4. $\partial b/\partial \theta_j = e^{\theta_j}/(1 + \sum_h e^{\theta_h}) = \pi_j = \mathbb{E}(y_j)$, as for any exponential family.

Hence the baseline-category logit model (7.1), which makes each θ_j linear in $\mathbf{x}$, uses the canonical link of this multivariate family: it is the multivariate analogue of logistic regression as a GLM with canonical link, and the sufficient statistics are $\sum_i y_{ij}$ and $\sum_i x_{ik} y_{ij}$. ■

Exercise 7.27 (A nonmonotone category probability)

Let $J = 3$ and

$$\pi_3(x) = \frac{1}{1 + \exp(\alpha_1 + \beta_1 x) + \exp(\alpha_2 + \beta_2 x)}.$$

Show that $\pi_3(x)$ is
(a) decreasing in x if $\beta_1 > 0$ and $\beta_2 > 0$;
(b) increasing in x if $\beta_1 < 0$ and $\beta_2 < 0$;
(c) nonmonotone when β_1 and β_2 have different signs.

Solution. Let $D(x) = 1 + e^{\alpha_1+\beta_1 x} + e^{\alpha_2+\beta_2 x} > 1$, so $\pi_3 = 1/D$ and $\pi_3' = -D'/D^2$ with

$$D'(x) = \beta_1 e^{\alpha_1+\beta_1 x} + \beta_2 e^{\alpha_2+\beta_2 x}.$$

(a) If $\beta_1, \beta_2 > 0$ then $D' > 0$, so $\pi_3' < 0$ everywhere.

(b) If $\beta_1, \beta_2 < 0$ then $D' < 0$, so $\pi_3' > 0$ everywhere.

(c) Say $\beta_1 > 0 > \beta_2$ (the other case is symmetric).

Step 1. $D''(x) = \beta_1^2 e^{\alpha_1+\beta_1 x} + \beta_2^2 e^{\alpha_2+\beta_2 x} > 0$, so D' is strictly increasing.

Step 2. As $x \to +\infty$ the first term of D' tends to $+\infty$ and the second to 0; as $x \to -\infty$ the first tends to 0 and the second to $-\infty$. Hence D' changes sign exactly once, at some x_0.

Step 3. Therefore $\pi_3' > 0$ for $x < x_0$ and $\pi_3' < 0$ for $x > x_0$: π_3 rises to a maximum at x_0 and then falls (with $\pi_3 \to 0$ at both ends). In particular it is not monotone.

This is the phenomenon noted after the alligator example: with $J > 2$, a category whose odds against the others move in opposite directions has a nonmonotone probability. ■

Exercise 7.31 (Cumulative logits need a common effect)

Consider logit $P(Y \le j \mid x) = \alpha_j + \beta_j x$, $j = 1, \dots, J-1$, with x ranging over $\mathbb{R}$ and the β_j not all equal. Show that the cumulative probabilities are misordered for some x, so the model does not define a probability distribution for all x.

Solution.

Step 1. Since the β_j are not all equal, there is a j with $\beta_j \neq \beta_{j+1}$ (otherwise all consecutive slopes, hence all slopes, would coincide).

Step 2. The difference of the two logits,

$$g(x) = (\alpha_j + \beta_j x) - (\alpha_{j+1} + \beta_{j+1} x) = (\alpha_j - \alpha_{j+1}) + (\beta_j - \beta_{j+1})x,$$

is a nonconstant linear function, so $g(x) > 0$ for all x on one side of the root $x^* = (\alpha_{j+1} - \alpha_j)/(\beta_j - \beta_{j+1})$.

Step 3. The logistic cdf F is strictly increasing, so for such x, $P(Y \le j \mid x) = F(\alpha_j + \beta_j x) > F(\alpha_{j+1} + \beta_{j+1} x) = P(Y \le j+1 \mid x)$. This would make $\pi_{j+1}(x) = P(Y \le j+1 \mid x) - P(Y \le j \mid x) < 0$, which is impossible.

The curves for j and $j+1$ are logistic curves with different slopes, so they cross; this is the formal reason behind the remark on nonparallel curves in Section 7.2.4. With x restricted to a bounded range, the model can still be valid there, but it is not a model for all x. ■

Exercise 7.32 (Uniform association in cumulative odds ratios)

For an $I \times J$ table with ordinal Y, independent multinomial rows, and scores $x_i = i$ for the rows, consider

$$\text{logit}\, P(Y \le j \mid X = x_i) = \alpha_j + \beta x_i.$$

(a) Show that logit $P(Y \le j \mid X = x_{i+1})$ − logit $P(Y \le j \mid X = x_i) = \beta$, and that this is the log cumulative odds ratio of the 2×2 table formed by rows $i, i+1$ and the binary response $(Y \le j,\ Y > j)$.
(b) Show that the residual df equal $IJ - I - J$.
(c) Show that independence of X and Y is the special case $\beta = 0$.
(d) If βx_i is replaced by unordered row parameters μ_i (treating X as nominal), show that for rows a, b the log cumulative odds ratio equals $\mu_a - \mu_b$ for all $J-1$ cutpoints.

Solution.

(a) The difference is $(\alpha_j + \beta(i+1)) - (\alpha_j + \beta i) = \beta$. Writing $P_i = P(Y \le j \mid X = x_i)$, the logit difference is

$$\log \frac{P_{i+1}/(1-P_{i+1})}{P_i/(1-P_i)},$$

the log odds ratio of the 2×2 table with rows $i+1, i$ and columns $Y \le j$, $Y > j$. It equals β for every pair of adjacent rows and every cutpoint: the association is uniform, so the model is a uniform association model in cumulative odds ratios.

(b) Each row is a multinomial with J cells and $J-1$ free probabilities, so the saturated model has $I(J-1)$ parameters. The model has $J-1$ intercepts and one β, i.e. J parameters. The residual df are $I(J-1) - J = IJ - I - J$.

(c) If $\beta = 0$, then $P(Y \le j \mid X = x_i) = F(\alpha_j)$ does not depend on i for any j, so all rows have the same conditional distribution of Y: this is independence. Conversely, under the model, independence requires $\alpha_j + \beta x_i$ to be free of i, i.e. $\beta = 0$. Testing $\beta = 0$ is then a test of independence on one df, which uses the ordering of both variables.

(d) With logit $P(Y \le j \mid X = a) = \alpha_j + \mu_a$, the log cumulative odds ratio of rows a, b at cutpoint j is $(\alpha_j + \mu_a) - (\alpha_j + \mu_b) = \mu_a - \mu_b$, free of j. The rows are then stochastically ordered by their μ_i, as in Proposition 7.5(c). ■